

Asearcher Evaluation — 8B

step-75 checkpoints

2026-08-05 · Qwen3-8B-Base, 32-turn wiki-18 local RAG · 7,152 questions over 10 benchmarks · greedy decoding · scored by strict EM, sub-EM, and gpt-oss-120b judge (frozen prompt `asearcher_mbe_v1`)

1 Key findings

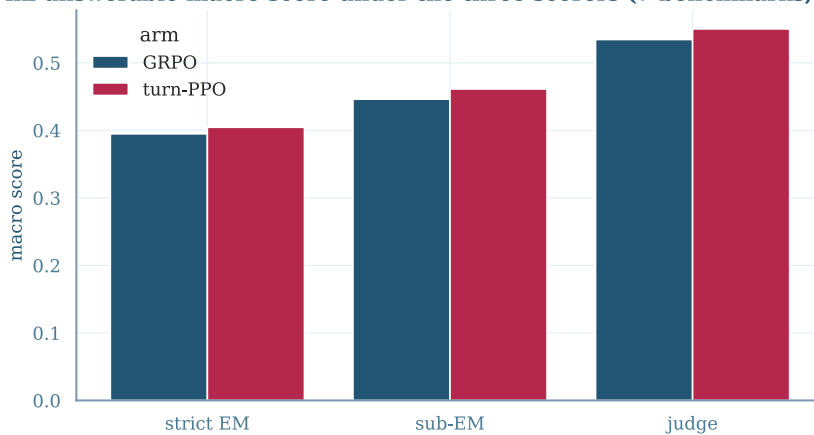
- **turn-PPO transfers ahead of GRPO, consistent with training val but with a much smaller margin:** wiki-answerable macro judge **0.550 vs 0.535** (strict EM 0.405 vs 0.395, sub-EM 0.461 vs 0.446). Training val said 0.561 vs 0.520.
- **turn-PPO reaches that score with 2–4× fewer turns** (4.3–12.3 turns/question vs GRPO's uniform 16.5–18.1) — same answer quality at a fraction of the search budget.
- **Strict EM (the training reward) understates real accuracy by ~0.13–0.14:** of 7,152 judged trajectories per arm, ~930–974 are judge-correct but EM-wrong, while only 3 are the reverse.
- **The live-web trio (GAIA / frames / xbench) floors at 0.05–0.23 judge** — a corpus gap (our retriever serves wiki-18 only), not a method difference.
- GRPO leaves **8.0%** of questions with no `<answer>` at all (loops to the turn cap); turn-PPO only 0.5%.

2 **Headline scores**

Both step-75 checkpoints evaluated over the full ASearcher test suite. Wiki-answerable = the 7 benchmarks our offline wiki-18 retriever can serve; live-web = the 3 that need the open web (reported separately — they measure the corpus gap, not credit assignment).

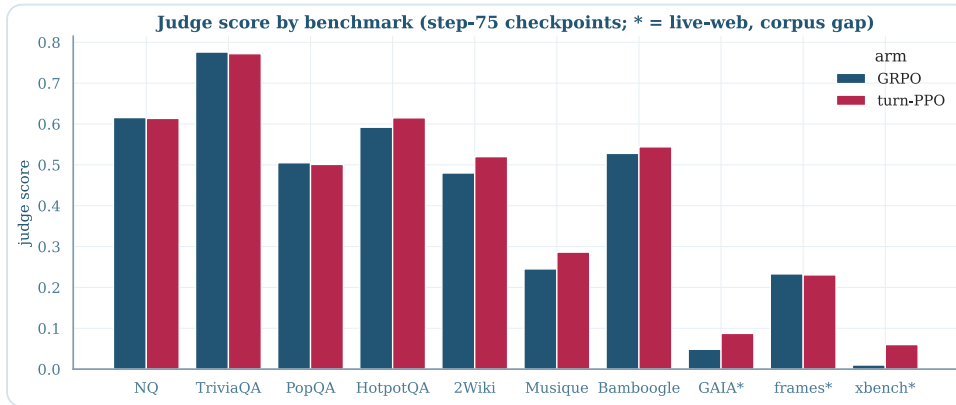
SLICE	ARM	N QUESTIONS	JUDGE	STRICT EM	SUB- EM
Wiki-answerable (7 benchmarks, macro)	GRPO	6,115	0.535	0.395	0.446
	turn-PPO	6,115	0.550	0.405	0.461
Live-web (3 benchmarks, macro)	GRPO	1,027	0.097	0.061	0.066
	turn-PPO	1,027	0.126	0.078	0.087
All 10 benchmarks (macro)	GRPO	7,142	0.403	0.295	—
	turn-PPO	7,142	0.423	0.306	—

Wiki-answerable macro score under the three scorers (7 benchmarks, 6,115 q)



Per-benchmark breakdown

Judge scores per benchmark. Single-hop (NQ / TriviaQA / PopQA) is a dead heat (± 0.01); turn-PPO wins every multi-hop benchmark (+0.02–0.04): HotpotQA 0.615 vs 0.592, 2Wiki 0.520 vs 0.480, Musique 0.286 vs 0.245, Bamboogle 0.544 vs 0.528.

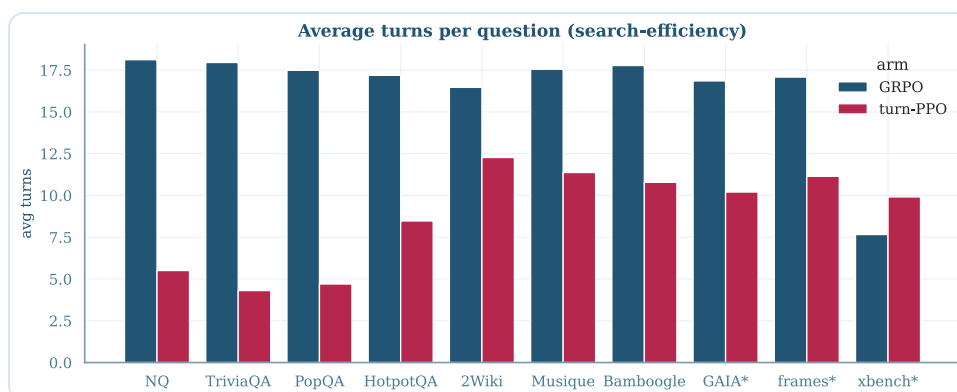


BENCHMARK	N	GRPO JUDGE	TURN- PPO JUDGE	GRPO EM	TURN- PPO EM	GRPO TURNS	TURN- PPO TURNS
NQ	999	0.616	0.614	0.428	0.433	18.1	5.5
TriviaQA	1000	0.776	0.772	0.577	0.572	18.0	4.3
PopQA	992	0.505	0.501	0.446	0.438	17.5	4.7
HotpotQA	1000	0.592	0.615	0.401	0.419	17.2	8.5
2WikiMultihopQA	1000	0.480	0.520	0.389	0.407	16.5	12.3
Musique	999	0.245	0.286	0.147	0.179	17.6	11.4
Bamboogle	125	0.528	0.544	0.376	0.384	17.8	10.8
GAIA live-web	103	0.049	0.087	0.049	0.078	16.9	10.2
frames live-web	824	0.233	0.231	0.135	0.125	17.1	11.1
	100	0.010	0.060	0.000	0.030	7.7	9.9

BENCHMARK	N	GRPO JUDGE	TURN- PPO JUDGE	GRPO EM	TURN- PPO EM	GRPO TURNS	TURN- PPO TURNS
xbench- deepsearch live-web							

4 Search efficiency

GRPO runs essentially every question to near the turn cap (16.5–18.1 avg turns, uniformly — including single-hop questions answerable in one search). turn-PPO modulates its budget with difficulty: ~4–5 turns on single-hop, 8–12 on multi-hop. At equal-or-better accuracy this is a **2–4× cheaper policy at inference time**, and matches the training-side observation (turn-PPO 6.5 searches/question vs GRPO 16.1 at step 75).



5 Scorer calibration

Three scorers on identical trajectories. Strict EM is the metric the training env rewards (`compute_score`) and what every `val/em` curve reports; the judge (gpt-oss-120b, frozen MBE prompt) is the closest to human judgment.

ARM	JUDGE-EM AGREEMENT	JUDGE✓ EM X	JUDGE X EM✓	EM VS ENV REWARD	NO- ANSWER FRAC	JUDGE FAILURES
GRPO	0.870	930	3	1.000	0.080	0
turn- PPO	0.863	974	3	1.000	0.005	0

The disagreement is one-directional (99.7% of disagreements are judge-correct/EM-wrong): strict EM misses formatting variants, aliases, and paraphrases. `EM vs env reward = 1.000` confirms the pipeline recomputes exactly the reward the env assigned — the eval is internally consistent with training. Judge calibration on the manual smoke set: 0.983 vs EM on 60 hand-checked items, with the one systematic strictness note that the judge occasionally rejects a technically-gold answer with wrong granularity.

6 Method & provenance

- **Checkpoints:** `token_grpo_qwen3-8b-base_32turn_fast75_s0` and `turn_ppo_b0_qwen3-8b-base_32turn_fast75_s0` at step 75 (75 steps, unfiltered Base-35k, fast config: partial rollout 8/4 + dynamic token batching + chunked prefill).
- **Eval protocol:** `val_only` greedy rollouts, 32-turn cap, top-5 wiki-18 e5 retrieval, identical env to training; full trajectory dumps on HDFS (`logs/p8b_eval_{grpo,turnppo}_s75/rollout_eval_*.jsonl.zst`).
- **Scoring:** `scripts/asearcher/judge_rollouts.py` — dump → trajectory fold → gold join → concurrent judging (gpt-oss-120b, TP4, mxfp4) → per-benchmark aggregation. Summaries in `outputs/judge/*.summary.json` ; cross-arm table `outputs/judge/cross_arm_s75.md` .
- **Training context:** val curves (strict EM, greedy, 512 fixed questions): GRPO 0.195 → 0.227 → 0.520 → 0.520; turn-PPO 0.195 → 0.234 → 0.479 → 0.561 @ steps 0/25/50/75.

```
# reproduce the cross-arm table
python scripts/asearcher/compare_eval_summaries.py \
```

```
outputs/judge/grpo_s75.summary.json outputs/judge/turnppo_s75.summary.json \
--out outputs/judge/cross_arm_s75.md
```

7

Caveats & pending

Read before quoting.

- **Single seed.** One run per arm; the wiki-7 gap (judge +0.015) is small — seed replication is queued before any headline claim.
- **Unfiltered training data.** Both arms trained on Base-35k without the closed-book filter (~40% of unfiltered training successes need zero searches) — the absolute numbers carry a recall confound; the 4B filtered campaign inverted the arm ordering.
- **Live-web $\neq$ method comparison.** GAIA/frames/xbench need the open web; our retriever serves wiki-18 (2018 snapshot). Those columns measure the corpus gap.
- **An earlier (2026-08-04 morning) claim that transfer "reverses" training val was wrong** — based on a mis-computed turn-PPO macro (0.355). The judged full-dump numbers here supersede it.
- **Pending:** the three in-flight arms (hcapo_ans, token_ppo, gigpo) get the same eval + judge pass when they reach step 75; this report then extends to the full 5-arm estimator comparison.